

Define erosion, dilation, opening and closing of an image and describe their effects. Briefly describe the basic algorithms of morphological operation. Explain gray-scale morphology.

Erosion

Erosion reduces the size of objects in a binary image by removing pixels from boundaries.

Effect:

- Removes small noise and thin protrusions.
- Detaches connected objects.
- Erodes/shrinks boundaries of objects

Dilation

Dilation is the opposite of erosion it increases the size of objects.

Effect:

- Joins adjacent objects.
- Fills small holes and gaps.
- Grows/expands boundaries of objects.

Opening

Opening is a compound operation that involves erosion followed by dilation.

Effect:

- Generally smoothes the contour of an object.
- Breaks narrow isthmuses.
- Eliminates thin protrusions and small objects.
- Preserves shape and size of larger objects.

Closing

Closing is another compound operation that consists of dilation followed by erosion.

Effect:

- Fuses narrow breaks and long thin gulfs.
- Eliminates small holes.
- Fills gaps in the contour.
- Preserves overall shape of objects.

Basic Morphological Algorithms:

1.Boundary Extraction: Finds edges by subtracting the eroded version of a shape from the original .

$$(A - (A \ominus B)).$$

2.Hole Filling: Fills internal gaps by repeatedly growing a starting point inside the hole until it hits the shape's boundaries.

3.Extraction of Connected Components: Identifies whole objects by starting from a single pixel and expanding until the entire connected shape is filled.

4.Convex Hull: The smallest shape that completely encloses an object without any inward dents .

5.Thinning: Shrinks a shape to a minimal thickness by removing border pixels while preserving its basic structure.

6.Skeletonization: Reduces an object to its central "spine" or medial axis.

7.Pruning: Cleans up thinned images or skeletons by removing small, unwanted branches.

Gray-Scale Morphology

1.Erosion: Erosion darkens and shrinks bright regions. Each pixel is replaced by the minimum value under the structuring element. Removes small bright noise and makes bright objects thinner.

2.Dilation: Dilation brightens and expands bright regions. Each pixel is replaced by the maximum value under the structuring element. Fills small dark holes and grows bright regions.

3.Opening: Opening = erosion followed by dilation. Removes small bright noise. Smooths object

boundaries. Does not significantly change object size.

4.Closing: Closing = dilation followed by erosion. Fills small dark holes. Connects nearby bright regions. Smooths boundaries from inside.